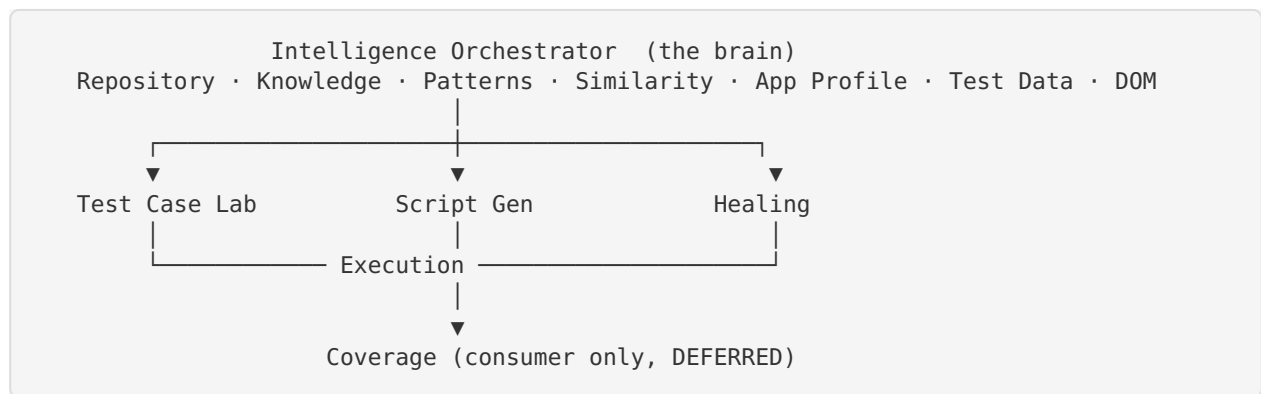


Intelligence Roadmap (revised per final review)

This supersedes the “Coverage Intelligence as master orchestrator” idea. Per the final review, **Coverage Intelligence is deferred and demoted to a pure consumer**. The **Intelligence Orchestrator remains the central brain** that every feature uses.

Architecture principle: **Rule-first → Intelligence-first → AI-last**.

Coverage observes, it never controls.



Phases (each = backend + frontend, shipped incrementally)

Phase	Scope	Status
1	Intelligence Score in the orchestrator (signature “grounded vs AI” metric)	✅ Done (backend)
2	Migrate Test Case Lab onto the orchestrator; expose <code>intelligenceScore</code> in <code>/generate</code> response	🕒 Next
3	Migrate Healing onto the orchestrator (evidence output + adapter)	🕒
4	Requirement Intelligence + Intent Intelligence layers feeding the orchestrator	🕒
5	Frontend : Intelligence Score UI across Script Gen, Test Case Lab, Healing	🕒
—	Coverage Intelligence dashboards / snapshots / trends / heatmaps / alerts / risk scoring	❌ Deferred

What each phase delivers

Phase 1 — Intelligence Score (this PR)

- New `IntelligenceScore` type on `OrchestratedIntelligence.metadata.intelligenceScore`:
`{ grounded, aiContribution, bySource, summary }`.
- `IntelligenceOrchestrator.computeIntelligenceScore()` — pure, deterministic, reusable by API + prompt block + dashboard.
- Rendered into the prompt block (“Intelligence Score: 94% grounded / 6% AI-generated”).
- Summary one-liner ready for UI: “94% grounded in repository intelligence. Only 6% AI-generated.”
- Fully additive — no behaviour change when the orchestrator flag is off.

Phase 2 — Test Case Lab migration

- `TestCoverageEngine.buildOrchestratedIntelligenceBlock()` (mirrors Script Gen).
- Sources: `['repository', 'appProfile', 'testData', 'knowledge', 'patterns']`.
- Skip legacy flat blocks when the orchestrated block is present (no double-injection).
- Return `intelligenceScore` in the `/api/test-coverage/generate` response.

Phase 3 — Healing migration

- Orchestrator gains method-index + RAG retrieval and an `evidence` output.
- `HealingIntelligenceContext` becomes a thin adapter over the orchestrator, preserving the `promptBlock` + confidence-boost `evidence` contract.

Phase 4 — Requirement + Intent Intelligence

- **Requirement Intelligence:** extract entities, workflows, actors, validations, dependencies, business rules before AI.
- **Intent Intelligence:** expand a requirement into intents (e.g. Login → Authentication / Positive / Negative / Boundary / Session / Security) so the repository search is intent-scoped and sharper.

Phase 5 — Frontend (dashboard repo)

- Intelligence Score badge/panel on generation results for all three features.
- Per-source breakdown bars; headline “% grounded / % AI-generated”.